Week 8 — Activity: Thermal Refraction in Layered Crustal Rocks

Geophysics 3A: Potential Fields & Geothermics

May 5, 2026

Purpose

To explore thermal refraction caused by conductivity contrasts within the crust, particularly in sedimentary and metamorphic layering.

Geological Context

Many crustal sections contain cyclic layering inherited from sedimentary environments, such as alternating shales and quartz-rich sandstones that later become quartzites.

- Shales: low thermal conductivity.
- Quartzites: high thermal conductivity.

This layering can persist through burial, deformation, and metamorphism.

Tasks

1. Consider a vertical sequence of alternating shale and quartzite layers. Assume constant heat flow through the sequence.
2. Using
$$q = -k\nabla T,$$
explain how the temperature gradient differs in shale layers compared to quartzite layers.
3. Sketch a qualitative temperature–depth profile showing changes in slope across layers.
4. Now consider the same layered sequence tilted relative to vertical heat flow. Describe how thermal refraction will distort isotherms.

Extension

- How can cyclic layering create an effectively anisotropic thermal conductivity?
- How might folding or tilting further modify the thermal field?

Connection

Relate this behavior to the physical-properties mixing demonstrations discussed earlier in the course.